

Grades will be notified at

7:30pm of Tuesday, July 2, 2024

**Question 1 (7.5 points)**

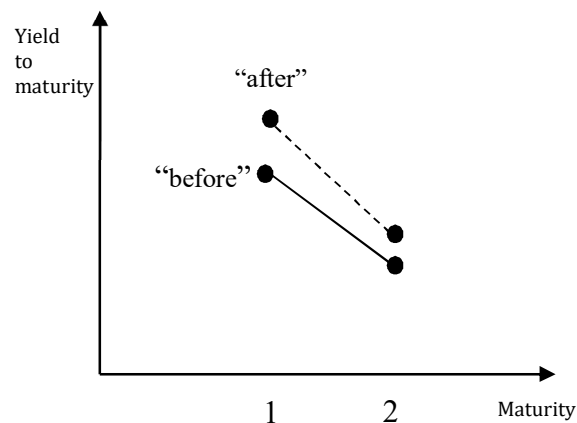
In country Alpha there are only one-year and two-year bonds. Current and expected future inflation are equal to zero, so nominal and real interest rates are the same. In period  $t + 1$  ("the future"), consumption depends only on disposable income at time  $t + 1$ , while investment depends on the values that interest rate and income will assume at time  $t + 1$ . As for the current period ( $t$ , "the present"), consumption depends positively on disposable income at time  $t$  and on the expected disposable income at time  $t + 1$ ,  $C_t = C(Y_t - \bar{T}_t, Y_{t+1}^e - \bar{T}_{t+1}^e)$ , while investment depends positively on current and expected future income levels, and negatively on the current and future expected levels of short-term interest rates. In both periods, the central bank chooses the interest rate, and  $G$  and  $T$  are exogenous as usual. Finally, at time  $t$ , in the initial equilibrium position both income and the interest rate take on positive values, while at time  $t + 1$  the economy is expected to be in a 'liquidity trap'. Suppose now that, at time  $t$ , the government announces that at time  $t + 1$  it will increase its purchases of goods and services. At the same time, the central bank announces that, at time  $t$ , it will adjust the policy rate so as to prevent any change in the equilibrium level of income for time  $t$ , and time  $t$  only ( $Y_t$ ). Explain how these fiscal and monetary policy announcements, which come as a surprise to individuals and are believed, will change the yield to maturity of the two-year bonds at time  $t$ ,  $i_{2t}$ .

**SKETCH OF ANSWER**

*The increase in  $\bar{G}_{t+1}$  will shift the IS for that period to the right, leading to an equilibrium characterized by a higher  $Y_{t+1}$  and an interest rate  $i_{1,t+1}$  still equal to zero (in fact, being the economy in a liquidity trap at time  $t + 1$ , the intersection between the new IS and the LM curve will continue to take place for an interest rate equal to zero). The higher expected future income will raise consumption and investment at time  $t$ , thus causing a rightward shift of the IS. To prevent  $Y_t$  from rising, the central bank will have to raise the time  $t$  policy rate. Since  $i_{2t}$  is (approximately) equal to the average of the short-term current and expected future interest rates, and that  $i_{1,t+1}$  remains equal to zero while  $i_{1,t}$  rises, the yield to maturity of the two-year bonds at time  $t$  will certainly increase.*

## Question 2 (7.5 points)

Using the information provided in the previous question, draw the yield curve prevailing in Alpha at time  $t$  before the central bank and government's announcements, and explain how it will be affected by those announcements. In particular, draw the new yield curve in the same graph, and explain if it is possible to determine without ambiguity how its slope and position will compare to those of the original curve.



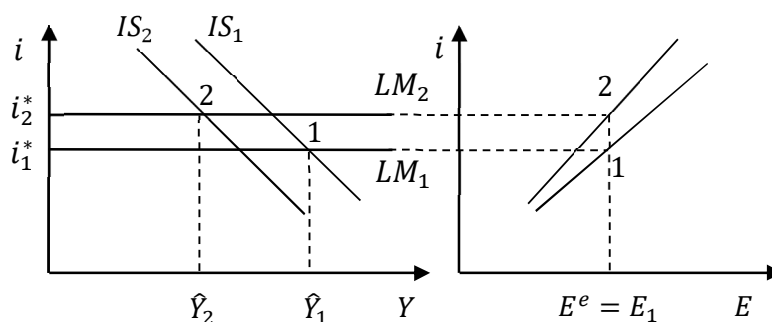
### SKETCH OF ANSWER

The initial yield curve is the continuous one in the figure. Its first point is the positive value the current short rate  $i_{1t}$  is initially taking on in this economy; being  $i_{2t} = \frac{1}{2}(i_{1t} + i_{1,t+1}^e)$  and  $i_{1,t+1}^e = 0$ , its second point is  $i_{2t} = \frac{1}{2}i_{1t} (< i_{1t})$ . The new curve (the dashed line in the figure) will lie above, and will be steeper than, the initial one. In fact, its first point will go up by  $\Delta i_{1t}$  (the amount by which the central bank will raise the policy rate at time  $t$ ) and its second one by  $\Delta i_{2t} = \frac{1}{2}\Delta i_{1t}$ , which is less than  $\Delta i_{1t}$ .

### Question 3 (7.5 points)

Consider an open economy with constant prices under a fixed exchange rate regime, in which net income from abroad is zero ( $NI = 0$ ) and investment is entirely exogenous ( $I = \bar{I}$ ). Aside from the exogeneity of investment, the economy is described by a standard “IS-LM-interest parity” model: the Marshall-Lerner condition is met, consumption depends on disposable income only, and  $G$  and  $T$  are exogenous. Represent the initial equilibrium of the economy in the pair of graphs below, denoting it by ‘1’, assuming that the foreign interest rate is initially  $i_1^*$ , the exchange rate equal to  $E_1$ , and that market participants expect that  $i$  will be kept at  $E_1$  also in the future, so that  $E^e = E_1$ .

Consider now an increase in the foreign interest rate, that becomes  $i_2^* > i_1^*$ , and a contemporaneous fall in foreign income,  $Y^*$ . Assuming that the central bank keeps the exchange rate fixed at  $E_1$ , explain if and how each of the curves depicted in the two graphs will be affected by these two contemporaneous changes. Represent the new equilibrium that the economy will reach, denoting it by ‘2’ in the two graphs, and explain if and why it is, or is not, possible to conclude that, in this new equilibrium, net exports ( $NX$ ) will be higher, lower, or unchanged.



#### SKETCH OF ANSWER

When  $i^*$  rises, in the second graph the straight line representing the interest parity condition rotates upwards and to the left; in the first one, the decrease in foreign income shifts the IS curve to the left. If the central bank kept the domestic interest rate at the initial level ( $i_1^*$ ), the exchange rate would depreciate. It follows that, to keep it fixed at  $E_1$ , the central bank will have to bring the domestic interest rate at the higher level the foreign one is taking on,  $i_2^*$ . The new equilibrium becomes ‘2’ in the figure. In the move from ‘1’ to ‘2’, net exports tend to rise for the decrease in domestic income, and to fall due to the decrease in foreign income. However, the latter force will prevail, so that net exports will be lower in the new equilibrium. In fact, in an open economy’s goods market equilibrium, national saving is equal to the sum of investment and net exports. But in this country investment is unchanged (it is exogenous), implying that net exports must change in the same direction, and by the same amount, as national saving. Being national saving lower in the new equilibrium (government saving is unchanged; private saving has gone down because of the decrease in  $Y$ ), the same must be true about net exports.

#### Question 4 (7.5 points)

At time  $t = 1$ , the economy of Country Beta is characterized by the following data:  $r = 15\%$ ,  $g = 10\%$ ,  $Y_0 = 60$ ,  $T_1 = 10$ ,  $G_1 = 10$ ,  $B_0 = 30$ . Compute the numerical value of  $Y_1$  – real GDP at time 1 – for this economy and, using the relation that gives the evolution of the debt-to-GDP ratio over time,  $B_1/Y_1$  – the value that, absent policy interventions, that ratio will take on at time 1. Finally, compute the value of the primary deficit-to-GDP ratio that, at time 1, would stabilize the debt-to-GDP ratio at the value it was taking on at time 0.

#### SKETCH OF ANSWER

*In this economy, one can compute time 1 GDP as  $Y_1 = (1 + g)Y_0 = (1 + 0.1) \cdot 60 = 66$ . Also notice that, in the same time period, the primary deficit-to-GDP ratio ( $d_1$ ) is zero. Plugging this values and the other available numerical information on this economy in the equation that gives the dynamics of the debt ratio, yields:*

$$\begin{aligned}\frac{B_1}{Y_1} &= (1 + r - g) \frac{B_0}{Y_0} \\ &= (1.05) \frac{30}{60} \\ &= 0.525\end{aligned}$$

*The value of the primary deficit-to-GDP ratio that, at time 1, would stabilize the debt ratio is the unique  $d$  that solves the following equation:*

$$\frac{B_1}{Y_1} - \frac{B_0}{Y_0} = (r - g) \frac{B_0}{Y_0} + d = 0,$$

*that is:*

$$d = (g - r) \frac{B_0}{Y_0} = (-0.05) \cdot 0.5 = -0.025.$$

*To stabilize the debt ratio in this economy, the government must run a primary surplus of 2.5% of the country's GDP, something that amounts to transforming the time 0 debt ratio into a steady state.*